

v4.3 — Mitosis Detection, Virtual HPFs, Mitotic Score (40×

E2E outcome: within confirmed hotspots, the system detects candidate mitotic figures at 40×, places 10 non-overlapping virtual HPFs over the densest region, and shows a live mitotic score (mitoses/mm² + classic per-10-HPF count). Every detection is toggleable; score recomputes in < 1 s on each toggle. The clinically hardest stage — biggest effort allocation.

Est: 2 weeks. Depends on v4.2 (confirmed hotspots). GPU appears here.

1. Scope

In: published-weights detector (+ optional published verifier), tiled 40× inference with cross-tile NMS, density kernel + greedy HPF placement, mm²-normalized scoring, detection review UI with live recompute. **Out:** training our own detector/verifier (future — pilot toggles become training data); HoVer-Net morphometry (v4.6 optional).

2. Models (published weights first — decision #6)

- **Detector:** MIDOG-trained published weights. Candidates in preference order: (a) MIDOG 2022 reference / winning entries with released weights (DeepMicroscopy repos), (b) published YOLO-family mitosis detectors trained on MIDOG++ . Selection task (day 1–2): run the 2–3 candidates on the MIDOG++ held-out test split, pick best F1; record scores in `models/detector/EVAL.md`. Wrap whatever wins behind one interface:

```
# pipeline/detect.py
```

```
class MitosisDetector(Protocol):
```

```
    def detect(self, tile_rgb: np.ndarray) -> list[Det]  # Det = (cx_px, cy_px, conf)
```

- **Verifier:** if the chosen pipeline publishes a companion classifier (several MIDOG entries do detector+classifier), use it: crops 128² @ 0.25 μm/px → `p_mitosis`. If none is usable, **ship detector-only** at a high-recall threshold and let the pathologist verify — the UI is built for it. Config flag `verifier.enabled`.
- Operating point: choose detector threshold on MIDOG++ val for recall ≈ 0.85–0.90 (recall-biased; misses are invisible to the reviewer, false positives are one click to remove). Document chosen thresholds in `configs/mitosis.yaml`.

- Serving: same worker container; `og-worker-gpu` Cloud Run service (nvidia-l4, 8 vCPU/32 GiB) handles stages `mitosis` only (poll filter by stage). PyTorch 2.x, fp16 inference, batch 16 tiles.

3. Inference pass (`worker/mitosis.py`)

1. For each confirmed, non-excluded hotspot polygon: enumerate $40\times$ tiles — 1024^2 px @ $0.25\text{ }\mu\text{m/px}$ ($256\text{ }\mu\text{m}$), stride 960 px (64 px overlap so borderline figures appear whole in ≥ 1 tile). Tiles clipped to polygon $\cap$ tissue mask.
2. Each tile: `read_region_srgb` → Macenko transform (same per-slide params from v4.1 — this is the domain-shift defense) → detector → optional verifier on crops.
3. Global NMS in μm space across tiles: greedy by confidence, suppress within $7.5\text{ }\mu\text{m}$ radius (a mitotic figure $\approx 10\text{--}15\text{ }\mu\text{m}$; duplicates come from tile overlap).
4. Persist per-candidate: centroid μm , `det_conf`, `ver_conf|null`, crop PNG (128^2 norm + 128^2 original) → GCS `artifacts/{case}/mitosis/crops/{id}.png`.
5. Runtime target: 50 mm^2 of hotspot ≈ 800 tiles → ≤ 8 min on L4 (measure; drop stride overlap if needed).

4. HPF placement + scoring (`pipeline/hpf.py`, pure functions)

4.1 Density map

- Grid $16\text{ }\mu\text{m}$; confirmed candidates (post-verifier threshold, pre-human) splatted as points; convolve with **disk kernel radius** `hpf.radius_um = 262` (config; $\triangleq$ spec's $250\text{--}270\text{ }\mu\text{m}$ HPF). Implementation: `scipy.signal.fftconvolve` with a precomputed disk mask.

4.2 Greedy placement (10 fields)

`mask = tumor_tissue` (union of confirmed hotspot polygons $\cap$ tissue mask)

for `k` in `1..10`:

`c_k = argmax(density)` over centers where:

`circle(c, r)` fully inside slide, center in mask,

`dist(c, c_j)  $\geq 2r$` for all placed `j` # non-overlap

place; zero density within $2r$ of `c_k`

fallback: if fewer than 10 legal positions (tiny tumor), place as many as fit;

scoring then uses the actual counted area (mm² normalization handles it —

this is exactly why decision #1 matters)

4.3 Scoring ([pipeline/scoring.py](#))

```
count_total = Σ mitoses inside placed HPFs      # after human toggles

area_mm2    = n_hpf * π * (radius_um/1000)2      # 10 × 0.2157 = 2.157 mm2 at r=262

density     = count_total / area_mm2

score       = lookup(configs/scoring.yaml)

# scoring.yaml default:

# mitotic_score:

# basis: per_mm2

# thresholds: {score2_min: 3.65, score3_min: 7.30} # Elston-Ellis 0-9/10-19/>=20 over 2.74
mm2

# display_classic_equivalent: true                # also show count/10HPF@0.274mm2 =
density*2.74
```

Pure, unit-tested exhaustively (boundary values, < 10 HPFs, zero mitoses).

5. Output JSON

```
{ "candidates": [

  { "id": "m_0001", "centroid_um": [123456.0, 78901.5],

    "det_conf": 0.91, "ver_conf": 0.87, "label": "mitosis",

    "crop_uri": "...", "crop_orig_uri": "..."}],

  "hpfs": [{ "seq": 1, "center_um": [...], "radius_um": 262, "count": 4}],

  "summary": { "count_total": 17, "n_hpf": 10, "area_mm2": 2.157,

    "per_mm2": 7.88, "classic_per_10hpf": 21.6, "mitotic_score": 3},
```

```
"model_versions": {"detector":"midog22_xxx@sha","verifier":null} }
```

6. Review UI (the most interaction-heavy screen — budget time here)

- Layers: HPF circles (numbered 1–10, count badge), detection markers (dot color by state: green confirmed-mitosis, gray rejected, amber unreviewed-candidate).
- **Gallery panel:** right-side grid of candidate crops sorted by confidence descending, keyboard-driven — **j/k** navigate, **m** = mitosis, **x** = not, **space** = jump viewer to the detection at 40×. Pathologists review 50–300 candidates; keyboard flow is the difference between 3 minutes and 15.
- Each toggle → optimistic local recompute (scoring logic duplicated in TS? **No** — **POST /stages/mitosis/recompute** with current label set returns summary; debounce 300 ms; server round-trip < 1 s satisfies the live requirement, one scoring implementation only).
- HPF editing: drag circle to move (non-overlap validated server-side on recompute), "re-place HPFs" button re-runs greedy placement using current human-confirmed labels as the point set.
- "Add missed mitosis": click at 40× → creates candidate **label=mitosis**, **source=pathologist_added**, crops cut server-side.
- Confirm gate: requires zero **unreviewed** candidates above **det_conf ≥ 0.5** OR explicit "accept remaining N as rejected" bulk action (logged).

7. DB additions

```
CREATE TABLE detections (
```

```
  id text NOT NULL, case_id uuid NOT NULL REFERENCES cases(id),
```

```
  hotspot_id text, centroid_um jsonb NOT NULL,
```

```
  det_conf real, ver_conf real,
```

```
  label text NOT NULL,      -- mitosis | not_mitosis | unreviewed
```

```
  label_source text NOT NULL, -- model | pathologist
```

```
  crop_uri text, PRIMARY KEY (case_id, id));
```

```
CREATE TABLE hpf_sites (
```

```
case_id uuid NOT NULL REFERENCES cases(id), seq int NOT NULL,  
  
center_um jsonb NOT NULL, radius_um real NOT NULL,  
  
mitotic_count int NOT NULL, source text NOT NULL DEFAULT 'model',  
  
PRIMARY KEY (case_id, seq));
```

Every label toggle → `audit_events(event_type='review_edit', payload={detection_id, from, to})`. **These toggles are the future training set — they are the product's moat; never lose one.**

8. Tests

- `hpf.py`: synthetic point sets → placement determinism, non-overlap invariant, fallback < 10 fields, density argmax correctness vs brute force on small grids.
- `scoring.py`: table-driven — (count, n_hpf) → expected score incl. boundaries (density exactly 3.65 → score 2).
- NMS: duplicated detections across synthetic tile overlap → single survivor.
- Detector smoke test: 3 MIDOG++ tiles with known annotations in CI (weights pulled from GCS, marked slow-test).
- Recompute endpoint: 300-candidate case → p95 latency < 1 s.

9. Acceptance checklist

- ☐ MIDOG++ held-out eval of chosen published detector documented (F1, precision, recall, threshold) in `models/detector/EVAL.md`; F1 ≥ 0.70 floor to ship (published models reach ~0.75+; below floor → escalate, don't ship silently).
- ☐ Full case run: hotspots → detections → 10 HPFs → score, ≤ 8 min GPU wall.
- ☐ Keyboard gallery review of 100 candidates in < 5 min by a non-pathologist tester.
- ☐ Toggle 5 labels → score updates < 1 s each; classic per-10-HPF equivalent displayed alongside per-mm².
- ☐ Tiny-tumor slide (< 10 HPFs fit) scores correctly via area normalization.
- ☐ All toggles present in audit trail with before/after.